

Information

General Remarks

- If asked for a single number, enter the number only, and omit the calculation or reasoning.
- If necessary, use a point to separate decimal places, e.g., 3.14.
- Wrongly formatted answers will be marked with a red triangle in the quiz navigation on the left (after clicking "Next page") and should be corrected before final submission.
- If anything is unclear, post in the Zoom chat, do not use the "Ask for help" button. We will either answer in the chat or ask you to come back to the main room (using the "Leave Breakout Room" link -- make sure not to leave the full meeting).
- Good luck!

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Bayesian Classification

Consider Bayes' Rule for classification:

$$P(\omega_i | \mathbf{x}) = \frac{p(\mathbf{x} | \omega_i)P(\omega_i)}{p(\mathbf{x})} \propto p(\mathbf{x} | \omega_i)P(\omega_i)$$

Consider a binary classification task over a set of classes $\Omega = \{\omega_1, \omega_2\}$, and answer each of the following questions in **one** sentence:

Question 1

Why can we say that $p(\mathbf{x})$ is irrelevant for determining the MAP class?

Comment:

Question 2

What will happen if the prior $P(\Omega)$ is very uneven (i.e., with a high value for some class, and a low one for other)?

Comment:

Question 3

Assume one particular class (say, ω_1) is so critical that we want a classifier that is guaranteed to classify all instances of class ω_1 as ω_1 , and does not miss a single one. How can we obtain such a classifier?

Comment:

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Consider the following two loss functions (cost matrices) (where λ_{ij} is the cost of predicting class i if the true class is j):

$$\lambda_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \lambda_2 = \begin{pmatrix} 0 & 10 \\ 10 & 0 \end{pmatrix}$$

Question 4

Do they lead to different values of the Bayes risk?

The correct answer is: yes

Question **5**

Do they lead a Bayesian classifier to make different predictions?

The correct answer is: no

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The k-Nearest-Neighbor Classifier

Which of the following statements is true (assume a numeric feature space, and Euclidean distance as our distance measure):

Question **6**

For $k = 1$, the k-NN classifier cannot overfit the training data.

The correct answer is 'False'.

Question **7**

Increasing k will increase the danger of overfitting.

The correct answer is 'False'.

Question **8**

Setting k to a high value will lower the bias of the classifier.

The correct answer is 'False'.

Question 9

If we add additional redundant features to the data (e.g., identical copies of some feature), the predictions of the k-NN classifier can change.

The correct answer is 'True'.

Question 10

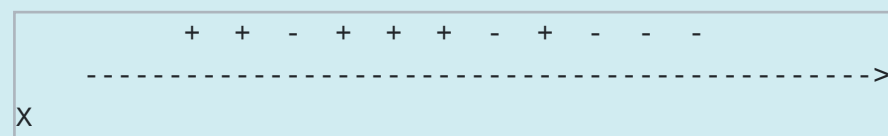
If we change the scale of one of the features (e.g., by multiplying all values by 1000), the predictions of the k-NN classifier can change.

The correct answer is 'True'.

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Decision Tree Learning

Consider the following simple one-dimensional (i.e., one numeric feature X only) training set in a two-class classification task:



Question 11

What is the minimum depth of a decision tree with binary splits on X that has an error of 0 on this dataset? (The depth of a tree that only consists of 1 node (the root) is 0; the depth of an arbitrary tree is the length of its longest branch).

The correct answer is: 3

Question **12**

Why can the same numeric feature be used several times along some path in a decision tree, but a symbolic feature can not?

Comment:

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Consider the following stop criterion for ID3: "If a node contains $\leq k$ examples, do not further split it; turn it into a leaf that predicts the majority class in these $\leq k$ examples."

Question **13**

Can this help against overfitting?

The correct answer is: Yes

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If we make k larger for this stop criterion, which of the following statements are true:

Question **14**

... the error on the training data cannot go down

The correct answer is 'True'.

Question **15**

... the error on independent test data cannot go up

The correct answer is 'False'.

Question **16**

... the error on independent test data cannot go down

The correct answer is 'False'.

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Classifier Evaluation

Consider a binary classification task over a set of classes $\Omega = \{\omega_1, \omega_2\}$.

Which of the following statements on precision and recall are true?

Question **17**

A classifier cannot have both Recall = 1.0 and Precision = 1.0 on class ω_1

The correct answer is 'False'.

Question **18**

A classifier cannot have both Recall = 1.0 and Precision = 0.0 on class ω_1 .

The correct answer is 'True'.

Question 19

If, on a specific dataset, a classifier A has a higher recall on class ω_1 than another classifier B, then A must have lower precision on ω_1 than B

The correct answer is 'False'.

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Which of the following statements on ROC analysis are true?

Question 20

No line segment of the ROC convex hull can be strictly horizontal.

The correct answer is 'False'.

Question 21

Classifiers in different places on the ROC convex hull can have the same classification accuracy.

The correct answer is 'True'.

Question 22

If two classifiers C1 and C2 have the same accuracy, then the difference between their TPRs ($TPR1 - TPR2$) must be exactly compensated by a corresponding difference in FPR: $(TPR1 - TPR2) = (FPR1 - FPR2)$.

The correct answer is 'False'.

Question **23**

If we multiply every entry in the cost matrix by 2.0, ...

The correct answer is: ... the optimal classifier will stay where it is

Question **24**

Would you rather be in ROC heaven or in ROC hell?

Comment:

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Neural Networks

Consider the following three possible loss functions for a regression problem, where

y_i is the true target value for input example x_i

$\hat{y}(x_i)$ is the value predicted by the model from input x_i :

- $L_1 = \sqrt{\frac{1}{|\mathcal{D}|} \sum_{x_i \in \mathcal{D}} (y_i - \hat{y}(x_i))^2}$
- $L_2 = \sum_{x_i \in \mathcal{D}} (y_i - \hat{y}(x_i))$
- $L_3 = \max_{x_i \in \mathcal{D}} (y_i - \hat{y}(x_i))^2$

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Which of these would be meaningful loss functions for a regression problem (and for a feed-forward neural network)?

Question **25**

L_1

The correct answer is 'True'.

Question **26**

L_2

The correct answer is 'False'.

Question **27**

L_3

The correct answer is 'True'.

Question **28**

Which of these three loss functions could be problematic for the backpropagation algorithm, and why? Answer in one sentence!

Comment:

Question **29**

When you add an additional hidden layer to a feed-forward network, how does that affect the *bias* of the network model?

The correct answer is: it decreases

Question **30**

And how does add an additional hidden layer affect the *variance* of the network model?

The correct answer is: it increases

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Deep Learning

Which of the following statements are true:

Question **31**

Deep learning researches the simulation of biological neurons with computers.

The correct answer is 'False'.

Question **32**

Artificial neural networks are mappings from tensors (vectors, matrices, or higher-order tensors) to other tensors.

The correct answer is 'True'.

Question **33**

The majority of parameters in a deep neural network is specified by hand.

The correct answer is 'False'.

Information

In the two-player game "Connect Four", players take turns inserting a chip of their player color (which is either red or yellow) into one of the board's seven slots, where it falls down to occupy the bottom-most free window. The game ends if one of the players manages to fill four connected windows (horizontally, vertically or diagonally) with their own color, or if the board is full.



You want to train a deep neural network to play the yellow player, by teaching it to look at the current board state (= network input) and suggest a move (= network output).

Question 34

How would you represent the board state for the neural network?

Comment:

Question **35**

How would you represent the expected output of the neural network?

Comment:

Question **36**

You are training it on recorded games of "Connect Four" experts. What is a possible data augmentation method you can use to derive additional training examples from the recordings? (Note that data augmentation is allowed to modify the targets.)

Comment:

Question **37**

Would you recommend only fully-connected (= dense) layers or also include convolutional layers for this task?

The correct answer is: Include convolutional layers

Question **38**

What was the worst / most difficult part of the ML group project?

Comment:

Question **39**

Was there anything you liked about it?

Comment:

Question **40**

I hereby declare that I did work on the exam independently and only the permitted aids were used, and, in particular, that no communication took place with a third party at any time during the examination.

The correct answer is 'True'.